

ORIGIN'26

Team SixSeven | Project: Campus ERP

Problem Statement: PS-6: ERP-based Integrated Student Management System

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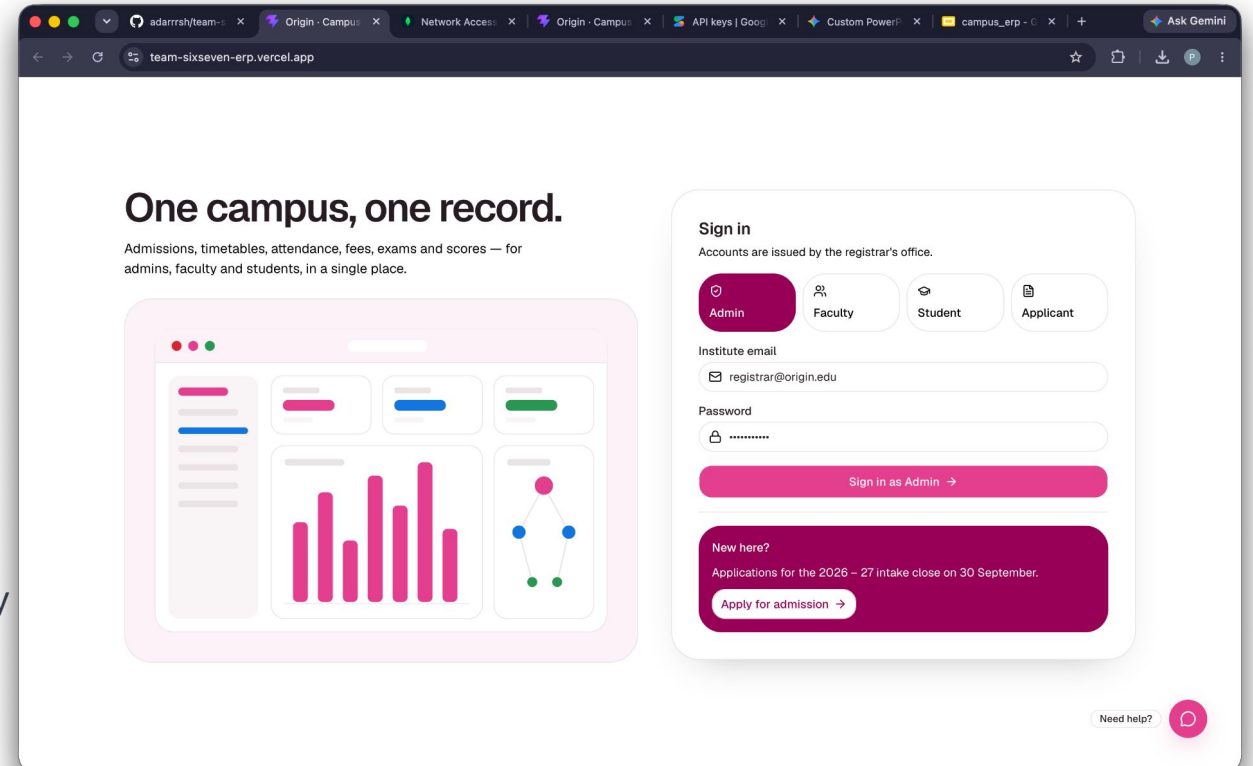
Deployment Link: <https://team-sixseven-erp.vercel.app/>

The Problem Statement (PS-6)

- × **Disconnected Systems:** Institutes run admissions, timetables, attendance, fees, exams and scores across disconnected spreadsheets, registers and paper trails.
- × **No Single Source of Truth:** The admin office, a faculty member and a student each see a different, often stale, picture of the same record.
- × **Manual Overheads:** Re-staffing a timetable when a teacher goes on leave is done by hand, class by class.
- × **Data Drift:** Marks are entered on paper before being typed into a spreadsheet, so grades and marks can silently disagree.
- × **Financial Disconnect:** Fee and fine balances are tracked separately from the ledger that produced them, so a due amount can drift from reality.
- ✓ **The Goal:** One ERP where every number is always the live truth, not a cached guess.

Proposed Solution

- A single web application with four role-based interfaces — Admin, Faculty, Student, Applicant — all reading and writing the same MongoDB-backed API.
- Nothing is hardcoded and nothing is pre-seeded: every figure (attendance %, fees due, grade, timetable slot) is computed live from the records that produced it.
- A background consistency checker continuously verifies that every derived number still agrees with its source records.
- A Gemini-powered chatbot on the login page guides new applicants through admission and sign-in.
- Force-directed knowledge graphs visualise how courses, faculty and students actually connect.



System Architecture

- ☰ One Express 5 REST API under /api, backed by a single MongoDB database via Mongoose 9.
- 📁 One React 19 single-page app, routed with React Router, that renders a different portal depending on which role signed in.
- 👤 Session identity is resolved from the signed-in user, not a hardcoded constant — the same component renders correctly for whichever student or faculty member is logged in.
- 🔗 Backend and frontend are two independent processes joined only by the API base URL, so either can be redeployed without touching the other.
- 🔍 A checkConsistency() audit function in the backend scans every collection and flags any figure that no longer matches its source.

React 19 SPA

Dynamic rendering based on role identity

Express 5 API

Independent process handling routes & consistency

MongoDB

Single live source of truth

Tech Stack



Frontend

React 19, React Router, Vite, Tailwind CSS v4, shadcn/ui, Framer Motion, React Flow (org chart), react-force-graph-2d (knowledge graphs).



Includes xlsx for spreadsheet export



Backend

Node.js, Express 5, MongoDB, Mongoose 9, Google Gemini API (chatbot), and Node's built-in test runner (node --test) for automated testing.



Data Strategy

No seed script. The database starts empty and every record is created through the app itself, keeping the demo data identical to what a real deployment would look like.

Login & Role-Based Access

- One login page serving four portals — Admin, Faculty, Student, Applicant — selected with a single toggle.
- The backend, not the frontend, decides which portal an account belongs to, so a faculty login can never land on the admin console.
- Session state persists in the browser so a refresh keeps the same signed-in identity.
- The same page carries the AI admissions chatbot for anyone who isn't sure how to apply or sign in.
- "Apply for admission" links straight into the applicant flow for new users.

Sign in

Accounts are issued by the registrar's office.



Admin



Faculty



Student



Applicant

Institute email

✉ registrar@origin.edu

Password

🔒

Sign in as Admin →

New here?

Applications for the 2026 – 27 intake close on 30 September.

Apply for admission →

Admin Portal: Dashboard & Organisation

 **Export to .csv / .xlsx** Every data table on every admin page can be exported directly to a spreadsheet with a single click.

-  Institute-wide dashboard: live counts of students, faculty and departments straight from the database (currently 9 students, 6 faculty, across 4 departments).
-  A React Flow organisation chart lays out the institute's actual department and reporting structure.
-  Departmental breakdowns and enrolment charts replace what used to be a static summary page.

Dashboard

Admissions

Faculty management

Student management

Finances

Time-table

Examinations

Score

R

↔

Dashboard

Everything the registrar's office needs to look at first thing in the morning.

PENDING ADMISSIONS

4

awaiting decision

FEEES OUTSTANDING

₹1,62,000

across all heads

FACULTY ATTENDANCE

96%

6 on staff

EXAMS SCHEDULED

2

6 courses running

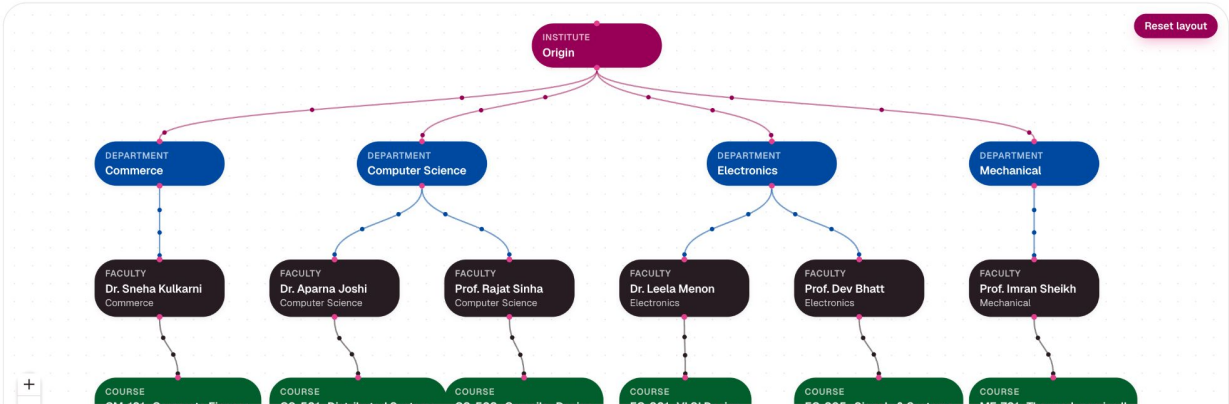
Review admissions →

Institute graph

Departments → faculty → courses. Drag any node; the graph is the source of truth for allocation.

Open faculty registry

Reset layout



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R

↔

Finances

Student fees, faculty payroll and the admission fee tracker, reconciled monthly.

TUITION COLLECTED

₹85,000

all heads

TUITION OUTSTANDING

₹1,62,000

2 overdue accounts

ADMISSION FEES

₹1,13,000

2026 – 27 intake

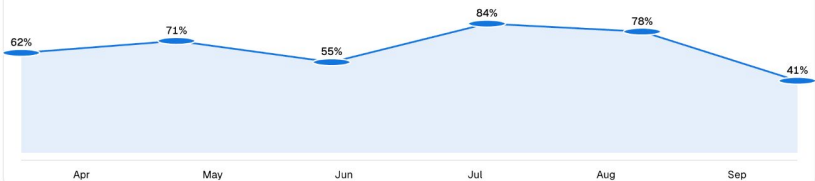
PAYROLL (NET)

₹6,95,700

Aug 2026

Collection rate

Percentage of the month's billing collected



Where the money sits

Current cycle

Tuition collected

₹85,000

Admission fees

₹1,13,000

Outstanding

₹1,62,000

Payroll committed

₹6,95,700

Student fees

Faculty salary

Admission fee tracker

Q

Search students or heads...

Export

STUDENT ID	NAME	PROGRAMME	HEAD	PAYABLE	PAID	BALANCE	DUE	STATUS
ST-8802	Vivaan Gupta	B.Tech CSE	Semester 5 tuition	₹42,000	₹0	₹42,000	2026-09-10	Overdue

Admin Portal: Admissions, Finance & Scores

- Admissions pipeline:** Review an application, approve/reject it, and collect the seat fee — approval and payment together issue the student a login.
- Finance dashboards:** Cover tuition collection, admission fees and payroll, each broken down by chart rather than a single number.
- Score overview:** Aggregates every published exam result across departments and courses.

A background job flags any paid admission that never resulted in an enrolled student, preventing lost payments.

4

awaitin

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R

↔

Decisions this intake

Pending, approved and rejected, out of 8 applications

8

total

Pending 4

Approved 2

Rejected 2

Requests 4

Approved

Rejected

Fee tracker

Q Search by name, programme or ID...

Export

APPLICATION	APPLICANT	PROGRAMME	ENTRANCE	APPLIED	EMAIL	PHONE	FEE PAYABLE	DECISION
AD-2048	Sara Fernandes	B.Tech CSE	89	2026-08-23	sara.f@mail.com	98200 21988	₹78,000	✓ Approve ✗ Reject
AD-2043	Aryan Mehta	B.Com Hons	71	2026-08-19	aryan.m@mail.com	98200 77310	₹52,000	✓ Approve ✗ Reject
AD-2042	Nandini Rao	B.Tech ECE	88	2026-08-18	nandini.r@mail.com	98200 55021	₹74,000	✓ Approve ✗ Reject
AD-2041	Ishaan Verma	B.Tech CSE	94	2026-08-18	ishaan.v@mail.com	98200 41190	₹78,000	✓ Approve ✗ Reject

Dashboard

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Score

R

↔

Score

Marks per student and exam. Editing here updates the student's portal immediately.

RECORDS 7 published marks

INSTITUTE AVERAGE 76% across every exam

DISTINCTIONS 3 85% and above

BELOW PASS 1 needs re-exam

Average by course

Across every published exam

80%

87%

58%

CS-501

EC-301

ME-701

Institute average

All published marks

76%

average score

Q Search students, exams or courses...

Export

STUDENT ID	NAME	PROGRAMME	EXAM	COURSE	MARKS	GRADE	UPDATE
ST-8801	Aisha Sheikh	B.Tech CSE	Mid-term · Sem 5	CS-501	88 / 100	A	Edit marks
ST-8807	Kavya Reddy	B.Tech CSE	Mid-term · Sem 5	CS-501	92 / 100	A+	Edit marks
ST-8802	Vivaan Gupta	B.Tech CSE	Mid-term · Sem 5	CS-501	61 / 100	C	Edit marks

Admin Portal: Timetable Management

-  **Shared Grid:** A single shared timetable grid — the exact same data students and faculty see, not a separate copy per role.
-  **Automatic Re-staffing:** Marking a teacher unavailable (or leave approved) automatically re-staffs their slots to a free colleague at read time.
-  **Seamless Recovery:** A returning teacher automatically gets their own slots back — nothing is permanently overwritten.

- Single-Teacher View:** Admins can pick one teacher at a time to see just their schedule, highlighted against the full grid.
-  **Audit Trail:** A "Changes" tab lists every reassignment made, flagging any slot that could not be covered as "Needs cover".
-  **Versioning:** The rebuild versions the timetable (v1, v2) so admins can see when it last regenerated.

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Time-table

Student and teacher timetables regenerate from teacher availability — mark someone unavailable and the grid re-staffs itself.

Rebuild - v1

Teacher availability

Click a teacher to mark them unavailable this week.

Dr. Aparna Joshi

Computer Science - free 3 slots

Dr. Leela Menon

Electronics - free 3 slots

Dr. Sneha Kulkarni

Commerce - free 3 slots

Prof. Dev Bhatt

Electronics - free 3 slots

Prof. Imran Sheikh

Mechanical - free 3 slots

Prof. Rajat Sinha

Computer Science - free 3 slots

View one teacher's timetable

Pick a teacher to see just their schedule in the Teacher view tab below — pick another to switch.

All teachers

Dr. Aparna Joshi

Dr. Leela Menon

Dr. Sneha Kulkarni

Prof. Dev Bhatt

Prof. Imran Sheikh

Prof. Rajat Sinha

Student view

Teacher view

Changes 0

Rooms and courses

What students see in their portal

PERIOD	MON	TUE	WED	THU	FRI
09:00 – 09:55	CS-501 A-204	EC-301 B-110	CS-503 A-206	Free	CS-501 A-204
10:00 – 10:55	CS-503 A-204	Free	ME-701 C-011	CM-101 D-002	EC-301 B-110

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Teacher view

Click a teacher to see just their schedule in the Teacher view tab below — pick another to switch.

All teachers

Dr. Aparna Joshi

Dr. Leela Menon

Dr. Sneha Kulkarni

Prof. Dev Bhatt

Prof. Imran Sheikh

Prof. Rajat Sinha

Student view

Teacher view

Changes 0

Rooms and courses

What students see in their portal

PERIOD	MON	TUE	WED	THU	FRI
09:00 – 09:55	CS-501 A-204	EC-301 B-110	CS-503 A-206	Free	CS-501 A-204
10:00 – 10:55	CS-503 A-204	Free	ME-701 C-011	CM-101 D-002	EC-301 B-110
11:10 – 12:05	Free	CS-501 A-204	EC-305 B-108	EC-301 B-110	CM-101 D-002
12:10 – 13:05	EC-305 B-108	CM-101 D-002	Free	CS-503 A-206	ME-701 C-011
14:00 – 14:55	ME-701 C-011	Free	CS-501 A-204	EC-305 B-108	Free

Faculty Portal

-  **Personal dashboard:** Scoped to the signed-in faculty member's own courses, students and weekly schedule. Includes attendance marking.
-  **Leave application flow:** The admin timetable reacts automatically the moment a leave is approved.
-  **Marks entry per exam:** Grades are computed automatically from marks — a faculty member never has to work out grade boundaries.
-  **Knowledge graph:** Force-directed graph showing the faculty member a

 Today

 My timetable

 Marks entry

 Network

 Leave

 Invigilation duty

 DA 

Today

Your classes, duty and attendance at a glance.

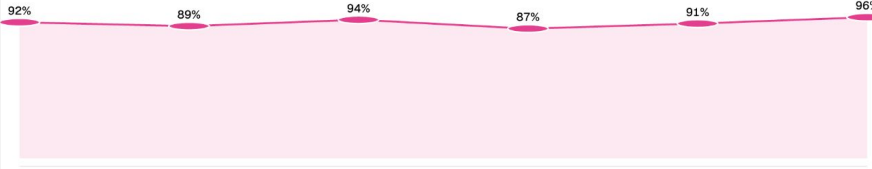
CLASSES THIS WEEK
4
CS-501

COURSES TAUGHT
1
62 students enrolled

MY ATTENDANCE
95%
rolling 30 days

INVIGILATION DUTY
0
assigned to me

My attendance trend
Monthly, as recorded by the registrar



My courses
Sem 5

CS-501

 Distributed Systems
62 enrolled · 4 credits

Next classes

Straight from the institute timetable

PERIOD	MON	TUE	WED	THU	FRI
09:00 – 09:55	CS-501 Dr. Aparna Joshi	EC-301 Dr. Leela Menon	CS-503 Prof. Rajat Sinha	Free	CS-501 Dr. Aparna Joshi
10:00 – 10:55	CS-503	Free	ME-701	CM-101	EC-301

 Today

 My timetable

 Marks entry

 Network

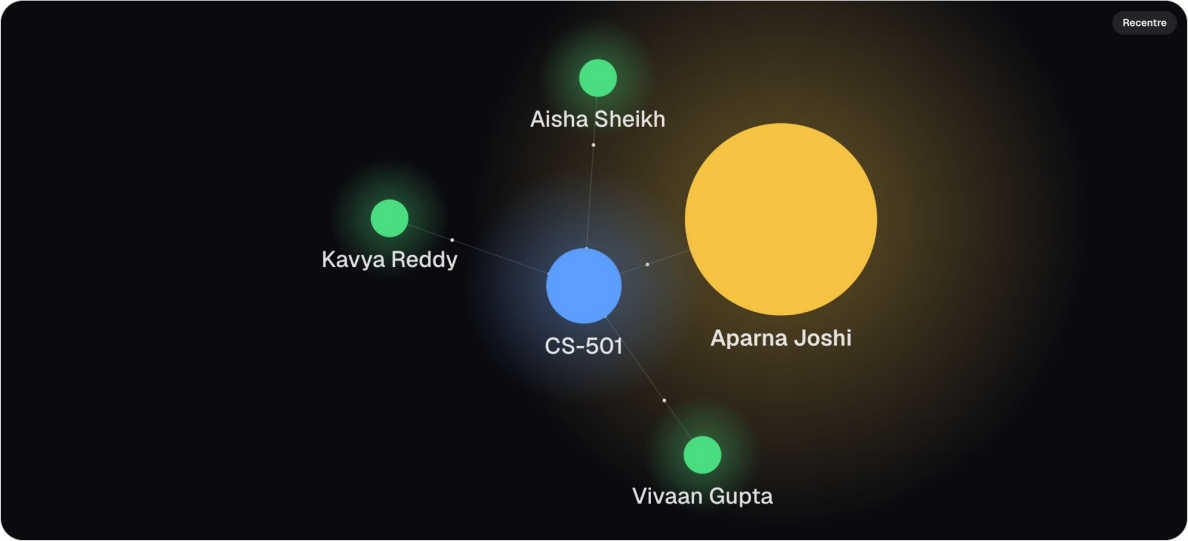
 Leave

 Invigilation duty

 DA 

Network

Courses I teach, and the students enrolled in each one — drag, scroll to zoom, click a node to focus.



Student Portal

-  **Personal dashboard:** Scoped to the student's own attendance, fees, fines, courses and scores.
-  **Progress visuals:** CGPA and attendance-trend charts replace a static percentage, making progress visible over time.
-  **Financial tracking:** Fee payments and outstanding fines with real backend errors surfaced instead of fake confirmations.
-  **Knowledge graph:** Force-directed graph putting the student at the centre, branching to enrolled courses and professors.

Attendance

Recorded when you tap in at the gate. 75% is needed to sit exams.

ATTENDANCE

95%

21 days recorded

DAYS PRESENT

20

tapped in

DAYS ABSENT

1

no record that day

CURRENT STREAK

20d

consecutive days present

Exam eligibility

You are clear to sit exams

95%

attended

Eligible

Recent days

Each bar is one day — full means present

100100100100100100100100100100100100100100100100100100100

09 Aug10 Aug11 Aug12 Aug13 Aug16 Aug17 Aug18 Aug19 Aug20 Aug23 Aug24 Aug25 Aug26 Aug27 Aug30 Aug31 Aug01 Sept02 Sept03 Sept05 Sept

Day-by-day

Your full attendance record

Q Search by date...

Export

DATE	STATUS	TAPPED IN	GATE
2026-09-05	Present	09:30	GATE-A

Network

My courses, and the professor teaching each one — drag, scroll to zoom, click a node to focus.

Prof. Rajat Sinha

CS-503

Aisha

CS-501

Dr. Aparna Joshi

Applicant Portal



Public Application Flow

A public application form for prospective students, entirely separate from the main sign-in flow to ease onboarding.



Live Status Tracking

Application status (Pending / Approved / Rejected) is visible to the applicant, driven directly by the same record the admin reviews.



Seat-Fee Gate

Seat-fee payment is only unlocked once approved. Paying this fee is the trigger that issues the applicant's actual student login.



Continuous Support

The login-page chatbot remains available throughout to answer "how do I apply" and "what happens next" questions instantly.

AI Admissions Chatbot



Gemini-Powered

Google Gemini-powered assistant embedded in the login page, scoped specifically to guiding visitors through the admission and sign-in process.



Structured Responses

Answers are capped at five lines and returned in a structured JSON format that the frontend renders directly for a clean UI.



Jailbreak Resistant

Highly resistant to jailbreak-style prompts ("ignore all previous instructions"). The bot stays on topic regardless of commands.



Graceful Fallback

Any question it can't answer, or any API failure, falls back to a fixed support contact (support@origin.edu) rather than hallucinating.

Data Integrity & Live Wiring

- ☰ **Zero Seed Data:** No hardcoded data anywhere in the app and no seed script — the database starts empty and every screen reads real records.
- 🔄 **Test Automation:** Backed by an automated test suite (Node's built-in test runner) asserting the database has zero consistency issues.
- 🕒 **Synchronous UI:** Every dialog waits for the real backend response before closing, showing real failures instead of optimistic fake successes.

Strict Cross-Validation: A background checkConsistency routine cross-validates:

- Attendance percentages vs register
- Fee balances vs fee heads
- Fine totals vs unpaid fines
- Grades vs raw marks
- Faculty leave status vs actual approved leave
- Every foreign reference against its target record

Conclusion & Future Scope

Origin replaces four disconnected spreadsheets with one live, role-aware ERP where every screen reads the same source of truth.

- ✓ **Delivered:** Admissions, timetabling with automatic re-staffing, attendance, fees, exams and scores, an AI admissions assistant, and knowledge-graph visualisations.
- **Future:** Real authentication and session tokens for enterprise security.
- **Future:** Integration of a production payment gateway.
- **Future:** Dedicated mobile app and richer historical analytics.

Deployment: <https://team-sixseven-erp.vercel.app/>